

Ideology

The bot's core design ideology can be summarized in three words:

Small. Efficient. Simplistic.

The bot will be designed to remain as small as possible without compromising the accuracy and reliability of its 3D-printed components. Every component and piece of equipment will be reduced to the bare minimum required to achieve the desired performance.

Weight will be treated as a valuable resource. Although the batteries and primary motor will contribute a significant portion of the overall mass, this is considered a necessary trade-off that will provide substantial benefits to the software and control systems.

Implementation of the Ideology

The three principles will be implemented throughout the robot's mechanical, electrical, and software design:

- **Low overall mass:** The target is to keep the total robot weight within approximately 100–200 g.
- **Minimal battery capacity:** The battery capacity will be limited to only what is required for reliable operation, reducing unnecessary weight.
- **Compact chassis:** The chassis will be kept as small as practical to reduce material usage, weight, and overall footprint.
- **Lightweight and maintainable software:** The code will prioritize simplicity and ease of debugging, using lightweight Python implementations supported by pre-existing and extensively documented libraries where appropriate.
- **Simplified CAD and mechanical design:** The robot will use simple geometry and a minimal number of components to reduce potential points of failure and improve reliability.
- **Nano-bot-specific design:** The mechanical design will account for the inherent differences between a full-sized robot and a highly compact robot, rather than directly scaling down conventional large-robot designs.

Overall, the design philosophy prioritizes achieving the required performance with the **minimum practical size, weight, complexity, and number of components**.